

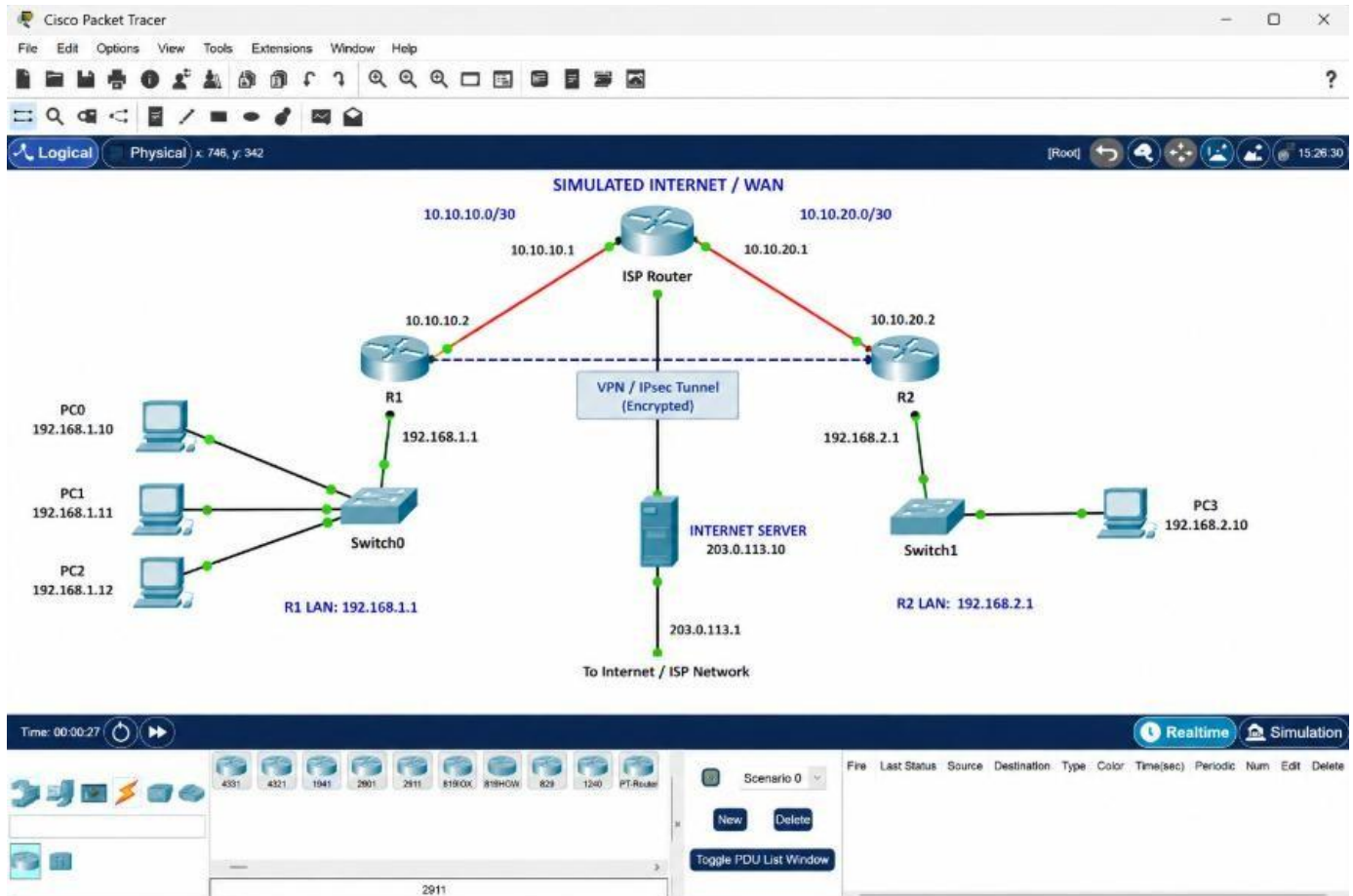
DAY-4

Weekly Practice and Troubleshooting Practical: Simulate a WAN setup with VPN and NAT. Task: Troubleshoot connectivity issues in the setup Checkpoint: Quiz: WAN technologies, VPNs, and NAT/PAT Practical Evaluation: Configure and troubleshoot a WAN.

[Q-1] Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.

(A)

- Create a simulated WAN connecting **two branch offices** through a VPN tunnel. Configure **PAT/NAT** so internal devices can access an external Internet server. Then intentionally create VPN and routing problems and troubleshoot them using show and debug commands.



▪ R1 Configuration

```
enable
configure terminal
```

```
hostname R1
```

```
interface gigabitEthernet0/0 ip
address 192.168.1.1 255.255.255.0
no shutdown
exit
```

```
interface gigabitEthernet0/1 ip address
10.10.10.2 255.255.255.252
no shutdown
exit
```

```
ip route 0.0.0.0 0.0.0.0 10.10.10.1
```

▪ R2 Configuration

```
enable configure  
terminal
```

```
hostname R2
```

```
interface gigabitEthernet0/0 ip  
address 192.168.2.1 255.255.255.0  
no shutdown  
exit
```

```
interface gigabitEthernet0/1 ip address  
10.10.20.2 255.255.255.252  
no shutdown  
exit
```

```
ip route 0.0.0.0 0.0.0.0 10.10.20.1
```

[Q-2] Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address. Hint: Use the 'overload' keyword in your NAT configuration.

(A)

Device	IP Address	Gateway
PC0	192.168.1.10	192.168.1.1
PC1	192.168.1.11	192.168.1.1
PC2	192.168.1.12	192.168.1.1

- **Configure NAT/PAT on R1**

enable

configure terminal

interface gigabitEthernet0/0

ip nat inside exit

interface gigabitEthernet0/1

ip nat outside exit

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet0/1 overload

end write

memory

- **PAT Result**

PC0 192.168.1.10
PC1 192.168.1.11
PC2 192.168.1.12

} — PAT Overload —→ 10.10.10.2

[Q-3] Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.

(A)

➤ **Configure GRE + IPsec VPN**

- Step 1: Create the GRE Tunnel ○ Step 2: Configure ISAKMP Phase 1
- Step 3: Configure IPsec

▪ **Router1**

```
enable
configure terminal
```

```
! =====
! STEP 1: GRE Tunnel Configuration
! =====
```

```
interface tunnel0 ip address 172.16.10.1
255.255.255.252 tunnel source
gigabitEthernet0/1 tunnel destination
10.10.20.2
no shutdown
exit
```

```
! Route to Branch 2 through GRE Tunnel ip
route 192.168.2.0 255.255.255.0 172.16.10.2
```

```
! =====
! STEP 2: ISAKMP Phase 1
! =====
```

```
crypto isakmp policy 10
encryption aes hash sha
authentication pre-share
group 2
exit
```

```
! Pre-shared Key
crypto isakmp key VPNKEY123 address 10.10.20.2
```

```
! =====
! STEP 3: IPsec Configuration
! =====
```

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
mode transport exit
```

```
crypto ipsec profile VPN-PROFILE
```

```
set transform-set VPN-SET exit
```

```
! Apply IPsec Protection to GRE Tunnel
```

```
interface tunnel0 tunnel protection ipsec
```

```
profile VPN-PROFILE exit
```

```
end write
```

```
memory
```

▪ Router2

```
enable configure
```

```
terminal
```

```
! =====
```

```
! STEP 1: GRE Tunnel Configuration
```

```
! =====
```

```
interface tunnel0 ip address 172.16.10.2
```

```
255.255.255.252 tunnel source
```

```
gigabitEthernet0/1 tunnel destination
```

```
10.10.10.2 no shutdown
```

```
exit
```

```
! Route to Branch 1 through GRE Tunnel
```

```
ip route 192.168.1.0 255.255.255.0 172.16.10.1
```

```
! =====
```

```
! STEP 2: ISAKMP Phase 1
```

```
! =====
```

```
crypto isakmp policy 10
```

```
encryption aes hash sha
```

```
authentication pre-share
```

```
group 2 exit
```

```
! Pre-shared Key
```

```
crypto isakmp key VPNKEY123 address 10.10.10.2
```

```
! =====
```

! STEP 3: IPsec Configuration

! =====

```
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac mode
transport
exit
```

```
crypto ipsec profile VPN-PROFILE
set transform-set VPN-SET exit
```

```
! Apply IPsec Protection to GRE Tunnel
interface tunnel0 tunnel protection ipsec
profile VPN-PROFILE exit
```

```
end write
memory
```

[Q-4] Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

(A)

➤ **Problem Scenario**

- Suppose PCs in Branch 1 can ping the Internet server: PC0 → 203.0.113.10 = SUCCESS
- But cannot ping Branch 2:
PC0 → 192.168.2.10 = FAILED

➤ **Troubleshooting Procedure**

□ **Step 1: Check Interfaces**

show ip interface brief Verify:

- LAN interface is up/up
- WAN interface is up/up
- Tunnel0 is up/up

- **Step 2: Check Tunnel Status** show
interface tunnel0
Verify the tunnel source and destination are correct.
- **Step 3: Check VPN Status** show
crypto isakmp sa show crypto ipsec
sa
If the security association is not established, investigate the ISAKMP/IPsec configuration.
- **Step 4: Check Routing**
 - On R1:
show ip route
 - The routing table must contain a route for:
192.168.2.0/24
 - The correct route is:
S 192.168.2.0/24 [1/0] via 172.16.10.2
 - If the route is missing, add:
configure terminal ip route 192.168.2.0
255.255.255.0 172.16.10.2 end
 - On R2, verify the return route: show ip route
The correct route should be:
S 192.168.1.0/24 [1/0] via 172.16.10.1
 - If missing:
configure terminal ip route 192.168.1.0
255.255.255.0 172.16.10.1 end
- **Main Configuration Change That Resolves the Issue**
 - The problem can occur when the branch-to-branch static routes are missing. ○ **Add on R1**
ip route 192.168.2.0 255.255.255.0 172.16.10.2
Add on R2 ip route 192.168.1.0 255.255.255.0
172.16.10.1
 - After adding both routes:

Branch 1 PC → R1 → GRE/IPsec Tunnel → R2 → Branch 2 PC Connectivity
should work.

[Q-5] Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it. Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.

(A)

➤ **Example Misconfiguration Found**

```
show running-config
show ip route show ip
interface brief
show crypto isakmp sa
show crypto ipsec sa
```

- The missing route was:
ip route 192.168.1.0 255.255.255.0 172.16.10.1
- I added the route and tested the connection again using:
ping 192.168.1.10
- After adding the correct return route, the branch-to-branch connectivity was restored.

➤ **Complete Troubleshooting Commands**

```
show ip interface brief
show interfaces tunnel0
show ip route show
running-config
show running-config | section crypto show
running-config | include crypto isakmp key show
crypto isakmp sa show crypto ipsec sa show ip
nat translations show ip nat statistics
show access-lists
```